CALIBER 2026 | Business proposal | Preliminary stage

Asset Knowledge Platform

A knowledge hub for the LLDPE unit. It links plant documents, work orders and live P&IDs to each equipment tag, and it answers engineers' questions with cited sources.

Connect — Data operations

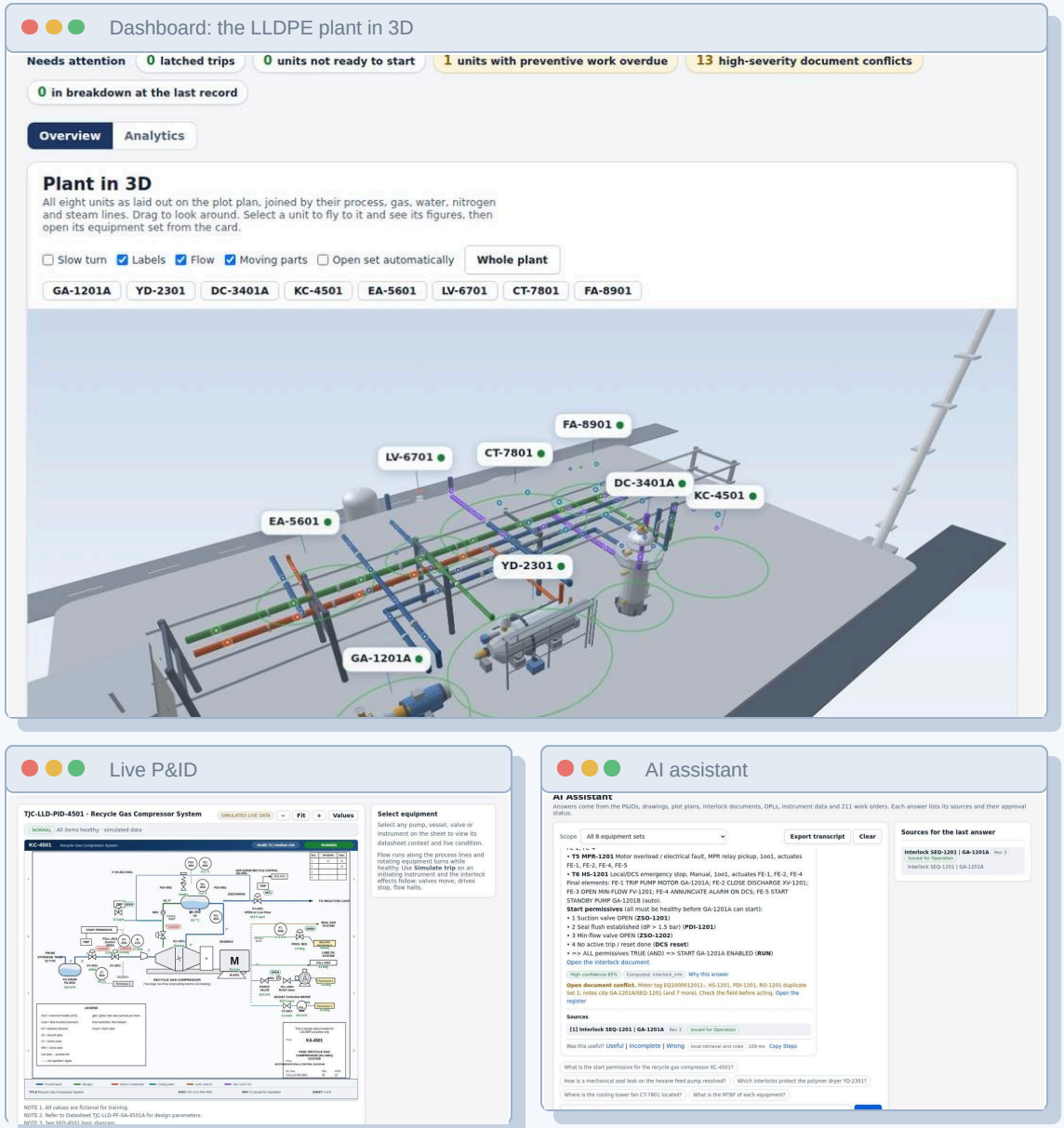
Ask — Verified answers

Act — Reliability and safety

564	Linked records	**1,109**	Searchable passages
20	Computed AI tools	**128/128**	Test questions passed

Chosen use case: Case 1, Manufacturing Knowledge Hub

Team: [Team name] **University:** [University] **Supervisor:** [Name]



Screens of the working prototype, built on the case data.

Problem: engineers cannot reliably find the right knowledge at the point of work, and this leads to improper execution

Root causes

Scattered sources. SOPs, P&IDs, datasheets, EDMS and AIMS records sit in separate systems.

Knowledge held by people. The reasons behind trips and repairs are not written down.

Conflicting documents. Revision and approval status are not visible where a document is used.

No link to the equipment tag. Work orders, drawings and interlocks cannot be searched together.

Core problem

Improper execution

Engineers start up, troubleshoot or repair equipment using a wrong step or an outdated drawing, without knowing a relevant past lesson.

Effects in the baseline data

8 of 55	Corrective jobs were repeated within 120 days.
434 h	Downtime in 31 breakdowns, with IDR 538 M of maintenance cost, of which 82% was corrective.
21%	Only this share of faults was found before it became a corrective job.

Who is affected, and how

Operators	Need the exact step and set point during start-up and trips.
Maintenance engineers	Need past repairs and spare parts for the same equipment.
Reliability engineers	Need a clean failure history to find repeat failures.
HSE and process engineers	Need current interlock logic and hazard limits.
Plant managers	Need to see which units are at risk and ready to start.

What the case data shows

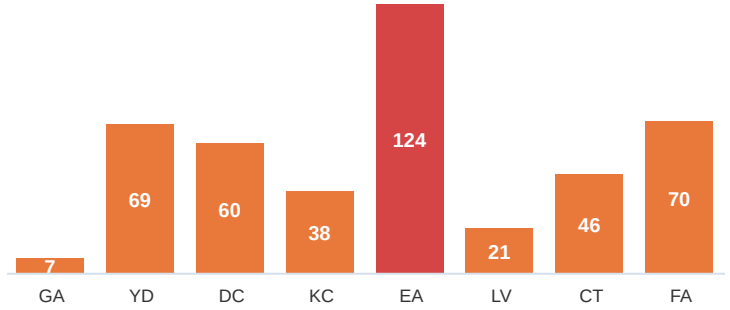
564
Records of 7 types, linked only through the equipment tag.

48
Document conflicts, for example FV-2302 is labelled both trips OPEN and trips CLOSED.

63
Failures in 211 work orders, none linked to a lesson learned.

624
Relations between records that the platform has to build.

Breakdown downtime by unit (hours)



Conclusion

Engineers lose time and make errors because plant knowledge is not linked to the equipment they are working on. In the LLDPE unit this shows up as 434 hours of breakdown downtime and repeated failures, so better access to existing knowledge has a measurable payoff.

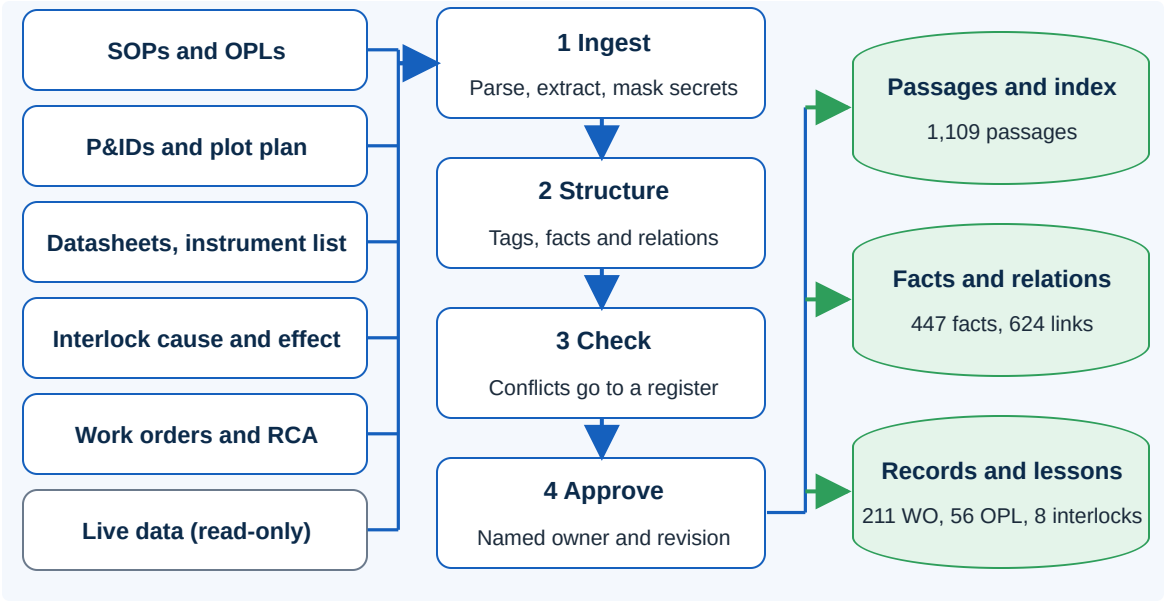
Source: CALIBER 2026 case book (Case 1). Baseline data from 8 equipment sets of the LLDPE unit, June 2024 to December 2025. Chandra Asri context: public releases on its Industry 4.0 programme and Cilegon digital twin

Solution: one checked knowledge base, linked by equipment tag, feeds an AI assistant that answers, predicts and plans

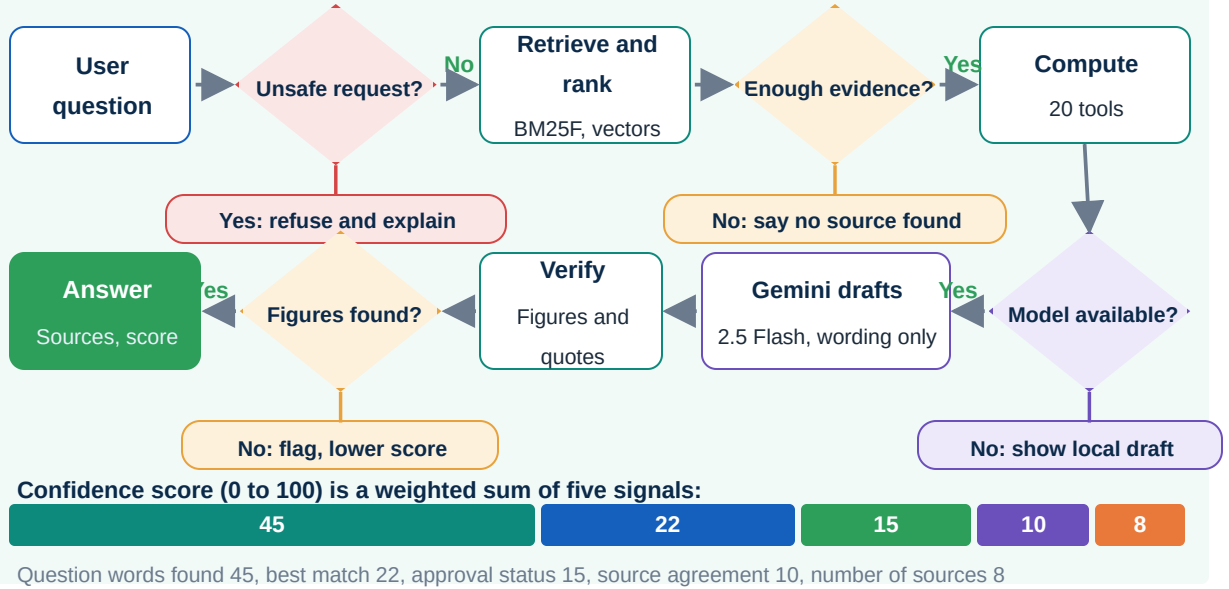
Objective: give every engineer one searchable, source-checked view of plant knowledge, linked to the equipment tag, and use it to answer questions, predict failures and plan work.

Key technologies: Gemini 2.5 Flash | BM25F and vector search | IndexedDB browser app | Netlify Functions and Blobs | PostgreSQL, pgvector (target)

Q1 Industrial Data Ops foundation: connect scattered plant knowledge



Q2 AI finds trusted information faster and reduces improper execution



Q3 Active use — Reliability, troubleshooting

- **Failure memory** — Similar cases and fixes
- **Predictive maintenance** — Risk, forecast, plan
- **OPL simulator** — Practise the lessons
- **Live P&ID, interlocks** — Trips and permissives
- **Start-up readiness** — Go or no-go check
- **Hazard and site plan** — Release, permits, replay

Conclusion

Site data enters through one checked pipeline and is linked to the equipment tag before anyone can search it. The assistant ranks the evidence first and uses the language model only to word the answer, and every figure is checked against its source.

Source: Prototype architecture and counts from the running application. Gemini 2.5 Flash is reached through a serverless proxy on the free tier

Solution: every question is routed to a computed function first, and the language model only words the verified result

User

- Operator
- Maintenance engineer
- Reliability engineer
- HSE engineer
- Plant manager

Example question

What usually causes a mechanical seal leak on the hexane feed pump, and how was it fixed?

AI assistant (in the browser)

1 Detect intent
Intent, tags and equipment

2 Safety guard
Unsafe requests are refused

3 Retrieve evidence
BM25F ranking and role filter

4 Choose function
Computed tool, or open question

Functions: 20 computed tools

Reliability and cost (7)
maintenance_stats, kpi_summary, failure_modes, predictive_maintenance, failure_memory, history_lookup, spares_outlook

Assets and limits (6)
equipment_info, spec_lookup, tag_facts, plot_plan_query, compare_equipment, rank_equipment

Operations and safety (6)
interlock_info, procedure_help, consequence_analysis, cascade_analysis, hazard_effects, safety_guard

Data quality (1): data_issues

Wording and checking

Computed result
Figures with their source records

Gemini 2.5 Flash words the answer
Free tier through /api/ai, temperature 0.2, up to 3 tool rounds. If it is unavailable, the local draft is shown.

Figures match the sources?

No: lower the score

Yes

Answer with sources and confidence

Worked example
Q2, from the prototype

Question	Intent	Function	Result	Answer
Seal leak on pump GA-1201A	Diagnosis from history	failure_memory	10 similar failures. Most supported fix, in 7 cases: replace the seal cartridge, clean the flush orifice, verify PDI-1201.	High confidence, 80%. Four sources, including WO-240002.

Conclusion

The assistant decides what to compute and what to retrieve before the language model sees anything. Gemini only words the result, and a figure that cannot be found in a cited source lowers the confidence score.

Source: Prototype assistant (intent router, retrieval index, 20 registered tools) and the Netlify function /api/ai. Tool names are the ones registered in the application

Business impact: at three units the platform could return about IDR 4.6 B a year and pay back in about 12 months

IDR 4.6 B
Value a year at 3 units

12 months
Payback at 3 units

IDR 7.9 B
Net benefit over 3 years

	1 unit (pilot)	3 units	6 units
Users	45	135	270
Value a year (IDR M)	1,521	4,562	9,123
One-off cost (IDR M)	1,400	2,100	3,150
Running cost a year	370	610	970
Payback	21 months	12 months	8 months
Net over 3 years	1,443	7,930	17,661

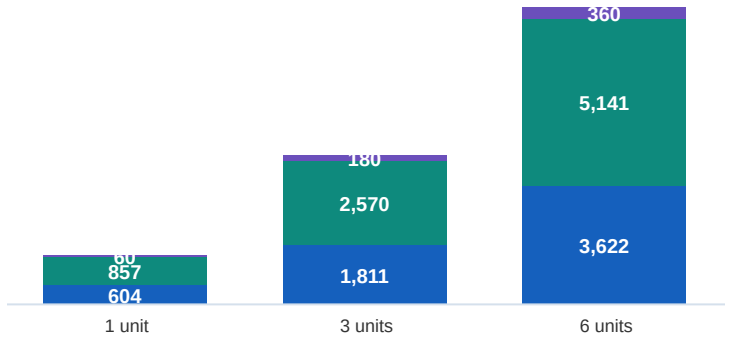
CASE DATA 20.4 breakdowns a year and a 14.0 h MTTR, from the 211 work orders.

ASSUMED 20 min saved per user a day, 230 days, IDR 175,000 per hour, IDR 30 M per downtime hour, 10% shorter repairs, IDR 60 M ramp-up.

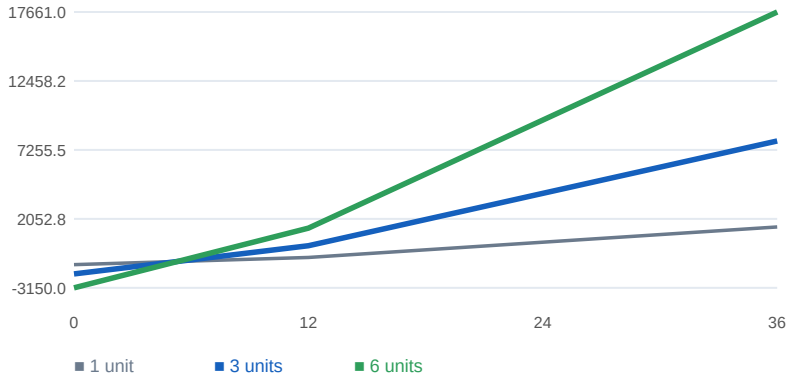
ESTIMATE Pilot cost of IDR 1,400 M one-off and 370 M a year (slide 6).

SCALING Each added unit matches the LLDPE unit. Shared cost IDR 1,050 M, data work 350 M per unit, support 120 M a year per added unit.

Annual value by lever (IDR M)



Cumulative net benefit (IDR M), months 0 to 36



KPIs and success metrics

KPI	Baseline	Target	Measured by
Time searching and checking sources	40 min a day (assumed)	20 min a day	Usage log and user survey
Mean time to repair	14.0 h	12.6 h (10% lower)	Work-order closure dates
Repeat failures within 120 days	8 of 55 jobs	30% fewer	Failure memory report
Answers with an approved source	Not measured	At least 90%	Answer audit log
Median time to a cited answer	About 15 min (assumed)	3 min or less	Timed pilot tasks
Users who rate answers useful	Not measured	At least 80%	Feedback buttons in the app

Conclusion Value grows with each unit while the shared platform cost stays fixed, so payback shortens from about 21 months for the pilot unit to about 12 months for three units. All scaling inputs are planning assumptions, and the pilot replaces them with measured Company data.

Source: Base inputs: 211 work orders (20.4 breakdowns a year, MTTR 14.0 h). All other inputs are assumptions or estimates, listed in the left panel, that the pilot replaces with Company data

Feasibility: a read-only pilot on the existing stack is realistic now, and each scale-up step is gated by a decision

Implementation feasibility: six readiness dimensions

Dimension	Status	In place today	Still needed
Technical	Ready	Prototype runs end to end; SQL schema and API specification are written.	Production database and ingestion queue.
Operational	Ready	Advisory only; works offline from local search; fits the OPL and work-order routines.	Shift-team trial and a champion in each function.
Legal	Decision needed	Drawings and vendor documents are used under their existing rights; data stays in region.	Enterprise AI terms with no training on Company data; PDP law review.
Cybersecurity	Planned	Roles, audit trail, secret masking, prompt-injection guards and rate limits.	Single sign-on, key vault, hosting outside the OT zone (IEC 62443), penetration test.
Data governance	Ready	Every record has an owner, revision and status; 48 conflicts are in a managed register.	Named data stewards and refresh rules for each source.
Organisation	Planned	A role-based approval workflow is built in.	Two-hour role-based training, monthly KPI review and an executive sponsor.

Deployment path and budget

Prototype today — Static web page, Netlify Functions and Blobs, Gemini 2.5 Flash on the free tier, local fallback.

Pilot, 90 days — Same stack with read-only EDMS and AIMS adapters, single sign-on and 45 users.

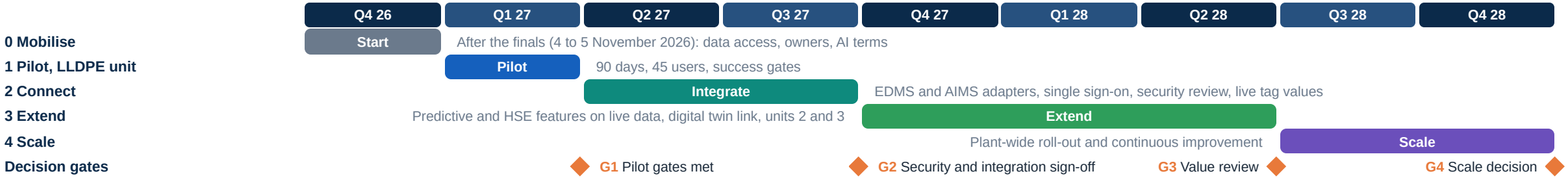
Production — PostgreSQL with pgvector, ingestion queue, a Company-approved model and SIEM export.

One-off budget of IDR 1,400 M and a running cost of IDR 370 M a year

Integration 650	Data 350	Change 250	150

Integration covers EDMS, AIMS and historian adapters. Data covers cleanup and stewards. Change covers training and champions. The last 150 is security: single sign-on, key vault and a penetration test. **Main risks and controls:** wrong answers (cited evidence and checks), poor data (conflict register), low adoption (champions), model outage (local fallback).

Implementation roadmap



Conclusion

The pilot needs only the existing stack and read-only exports from EDMS and AIMS. Four decision gates let the Company stop or continue at each step, and the pilot itself needs a one-off budget of IDR 1.4 B.

Source: Prototype and architecture notes. Dates and budget are proposals for discussion with the Company

Conclusion: a 90-day pilot will show whether the platform reduces search time and repair time at the LLDPE unit

Proposed solution

- Links 98 documents and 211 work orders to the equipment tag.
- Answers questions with cited sources and a confidence score.
- Supports failure analysis, simulation, start-up checks and planning.
- Uses a free-tier model with a local fallback, so the pilot needs no paid model licence.

Key value

IDR 1.5 B
Value a year for one unit, base case

21 months
Payback for one unit

IDR 4.6 B
Value a year at three units

20 min
Saved per engineer each day (assumed)

Expected outcomes

- At least 90% of answers carry an approved source.
- The median time to a cited answer is 3 minutes or less.
- At least 80% of users rate the answers as useful.
- No answer is given without a source.
- At least 90% of the 48 document conflicts are closed or accepted.

Key questions of the case, and how the proposal answers each

1	**How can the Company build a structured Industrial Data Ops foundation to connect scattered plant knowledge sources?**	Build a connected knowledge foundation. Checked ingestion links 98 documents and 211 work orders by equipment tag, places 48 conflicts in a register and requires an owner's approval.
2	**How can AI help engineers find trusted technical information faster and reduce the risk of improper execution?**	Improve search, Q&A and answer verification. Answers carry cited sources and a confidence score, figures are checked, and unsafe requests are refused (128 of 128 test questions passed).
3	**How can the Manufacturing Knowledge Hub be integrated with operational systems to support reliability, troubleshooting and continuous improvement?**	Active knowledge for decisions. Read-only links to EDMS and AIMS feed failure memory, a backtested forecast, start-up readiness and site planning.

Decision requested: a 90-day pilot

Weeks 1 to 2: Agree data access, owners and AI terms.

Weeks 3 to 6: Load the LLDPE unit and train 45 users.

Weeks 7 to 12: Measure against the gates, then decide.

Conclusion

The platform gives engineers cited answers at the point of work and gives the plant a basis for reducing repeat failures. We ask the Company to approve a 90-day pilot, so that the value is measured with Company data before any decision to scale.

Source: Prototype test results (128 of 128 evaluation questions; static, regression and layout checks) and the business case on slide 5

Team profile

Replace the bracketed fields with the team's details before submission

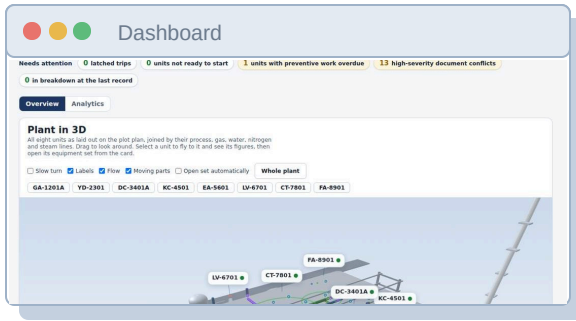
Team name: [Team name] **University:** [University] **Supervisor:** [Supervisor name]

No.	Name	Major	Semester	Area of expertise	Contribution to the proposal
1	[Name]	[Computer Science]	[6]	[AI and software]	[Formulated the idea; built the prototype and the AI assistant]
2	[Name]	[Chemical Engineering]	[7]	[Process and reliability]	[Problem analysis, process content and hazard models]
3	[Name]	[Industrial Engineering]	[7]	[Maintenance and business case]	[Business impact, KPIs and roadmap]

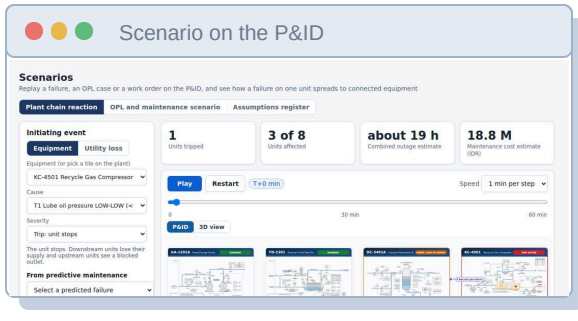
The booklet allows two to three members from the same university. Delete any unused row.

Screens of the working prototype

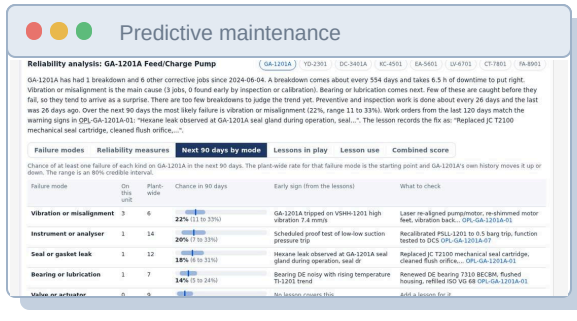
Dashboard



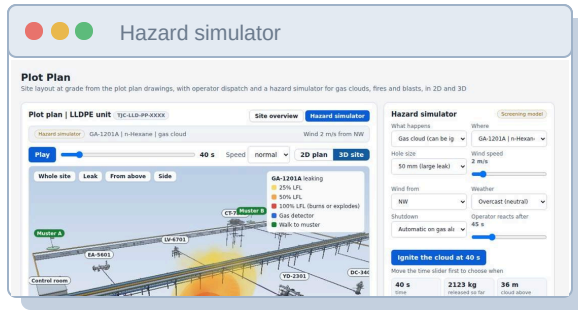
Scenario on the P&ID



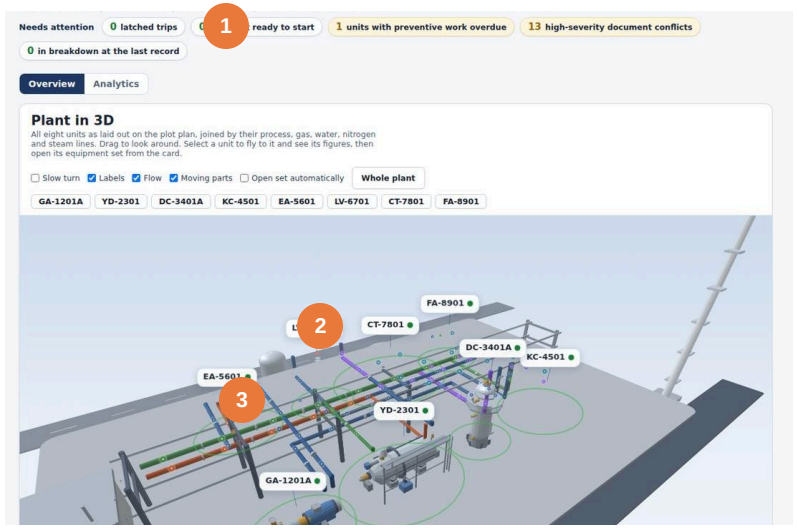
Predictive maintenance

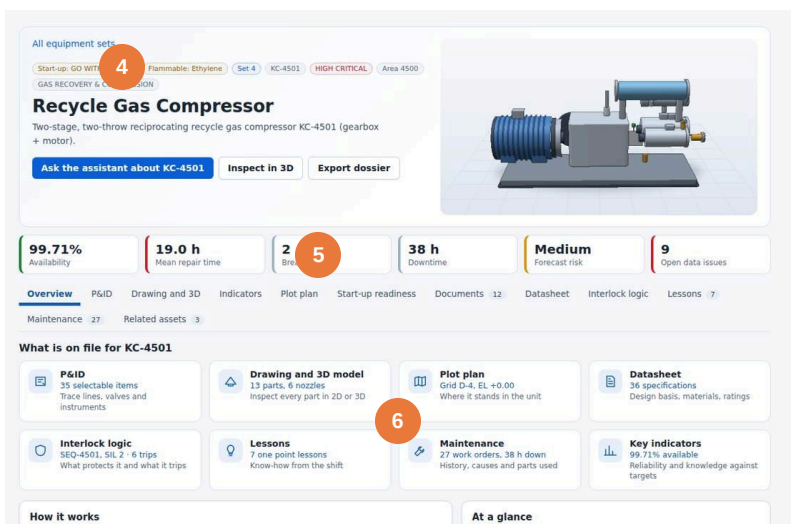


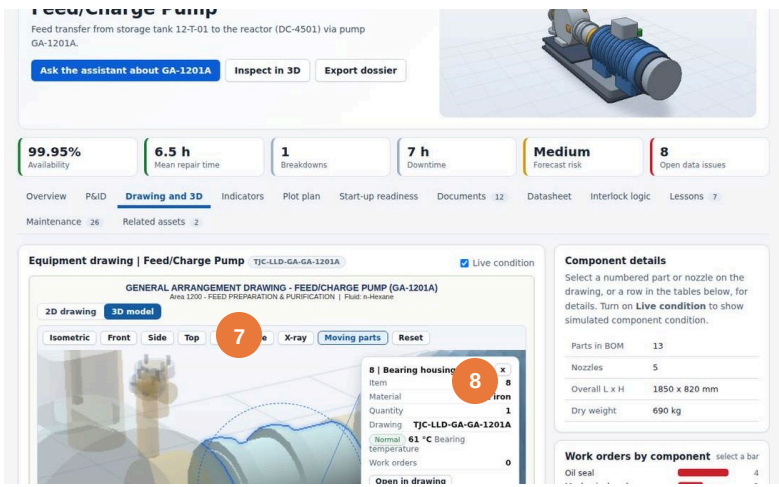
Hazard simulator



A1 Dashboard, equipment pages and the 3D plant







1. Chips count latched trips, units not ready, overdue preventive work and document conflicts.
2. Each unit has a status label. Selecting a unit flies the view to it.
3. Pipes are coloured by service: process liquid, nitrogen, steam, cooling water, lube oil and gas.
4. The equipment page opens with start-up status, criticality, flammability and a 3D model.
5. Key figures: availability, mean repair time, breakdowns, downtime and forecast risk.
6. Cards link to the P&ID, drawing, plot plan, datasheet, interlocks, lessons and maintenance history.
7. Each machine model is built from the datasheet bill of materials, with view and moving-part controls.
8. Selecting a part shows its item number, material, drawing and live condition.

Operators and managers

See the status of every unit and the work that is open without opening a report.

Maintenance engineers

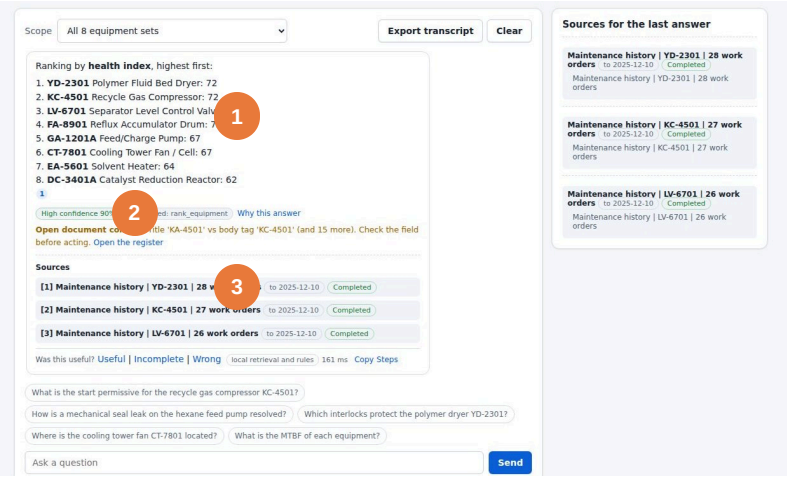
Find a part, its material and its drawing from the same page as the work-order history.

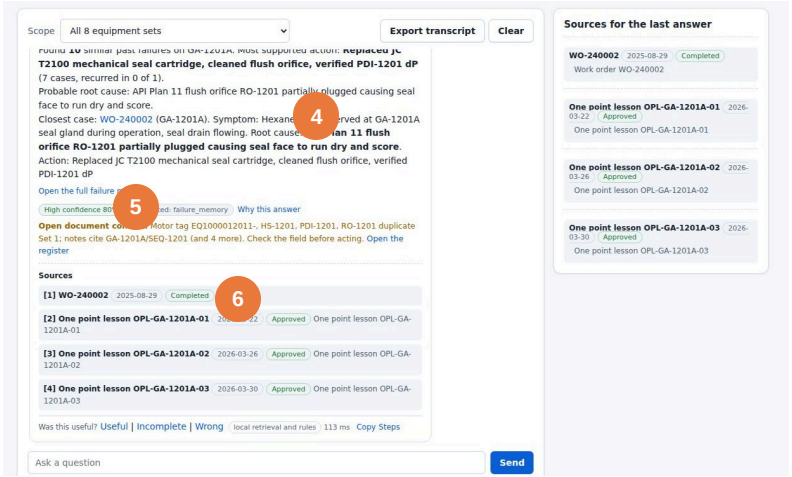
Reliability and HSE engineers

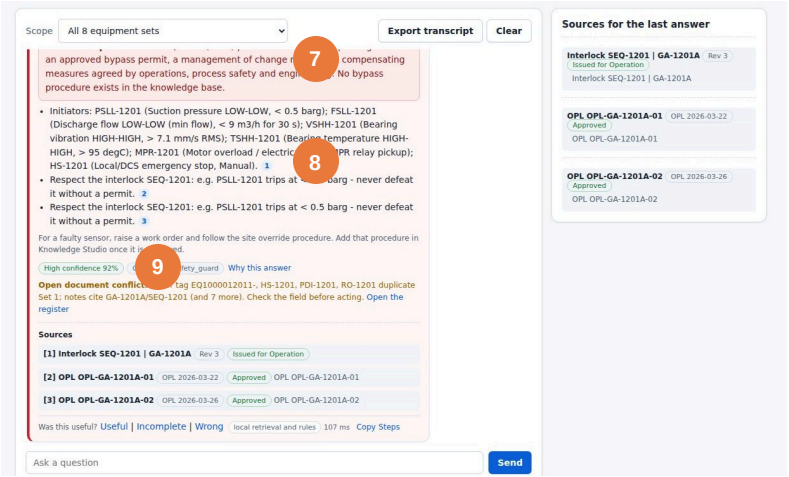
Open the interlock logic, lessons and failure history of a unit in one click.

Source: Prototype dashboard and equipment pages, built on the case baseline data

A2 AI assistant: three kinds of answer







1. Ranking by health index, calculated by the rank_equipment tool from the work-order history.
2. A confidence chip shows 90% and the tool that computed the figures.
3. Cited sources show their approval status and date.
4. Failure memory finds similar past failures and names the most supported action and root cause.
5. Confidence is 80%, and the answer links to the closest work order.
6. Sources are one work order and three approved OPLs.
7. A request to bypass a trip is refused, and the reply gives the approved route.
8. The reply still lists the initiators and set points of the interlock.
9. The refusal comes from the safety_guard tool, not from the language model.

Scope selector
Limits a question to one equipment set or to the whole plant.

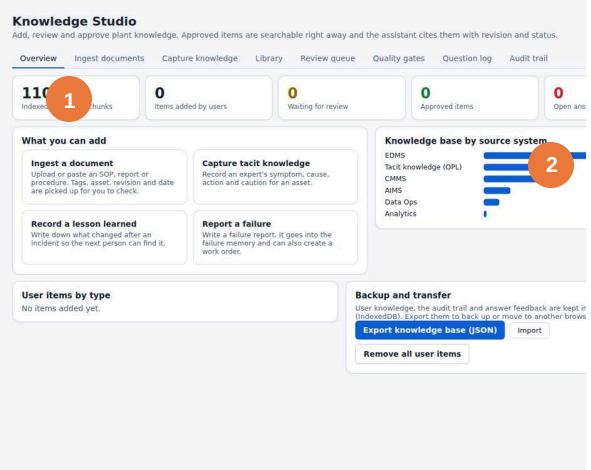
Suggested questions
Generated from the plant data and shown under each answer.

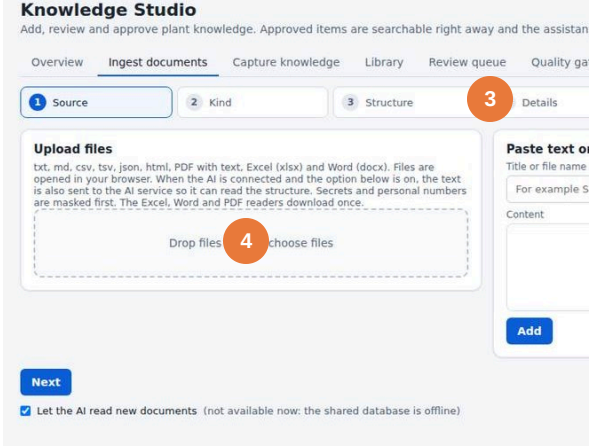
Open conflict note
Appears when an answer touches a document with a recorded conflict.

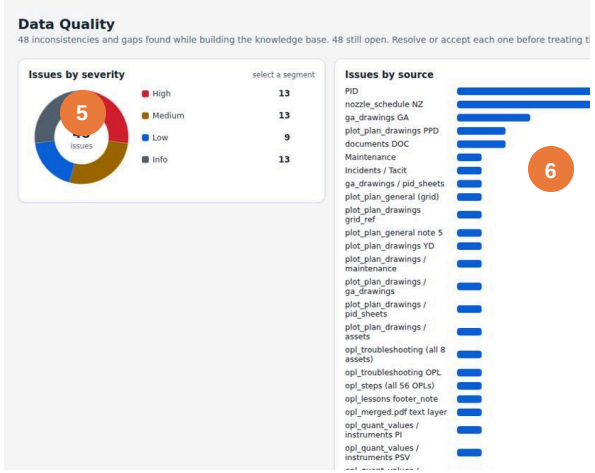
Feedback links
Useful, Incomplete and Wrong go to a review queue.

Source: Prototype AI assistant running on local retrieval and computed tools. The same screens appear with Gemini wording when the free-tier key is connected

A3 Knowledge base and data operations



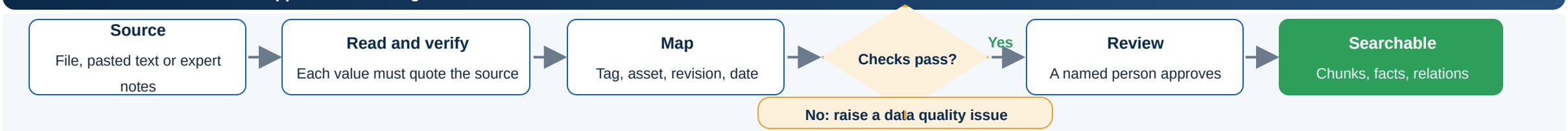






1. 1,109 indexed knowledge chunks.
2. Chunks by source system: EDMS 478, OPL 280, CMMS 211, AIMS 84.
3. Six steps: source, kind, structure, details, checks and create.
4. Files are read in the browser; secrets are masked before any AI extraction.
5. 48 inconsistencies found while the base was built, by severity.
6. Each issue is tracked by source until it is resolved or accepted.
7. The equipment tag sits at the centre of the graph.
8. Documents, OPLs, instruments, interlocks and work orders link to it.

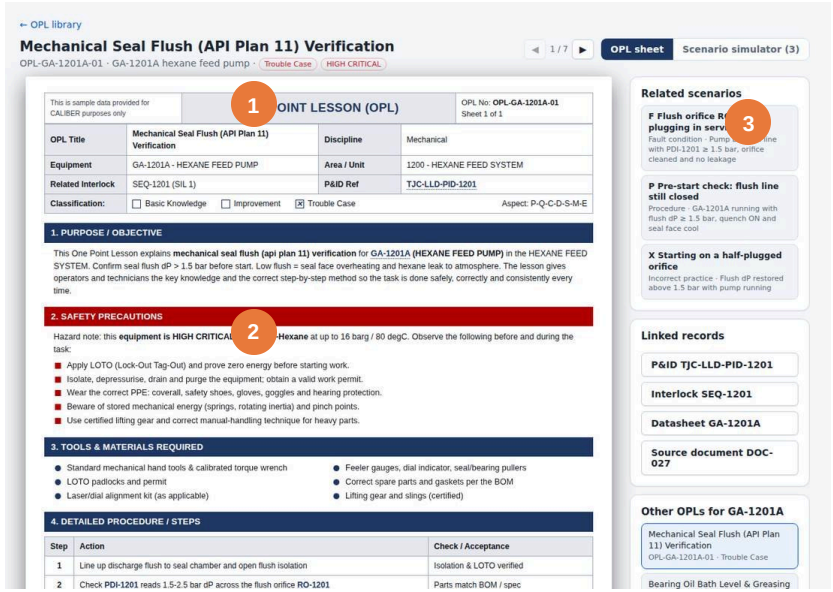
How a new document becomes approved knowledge



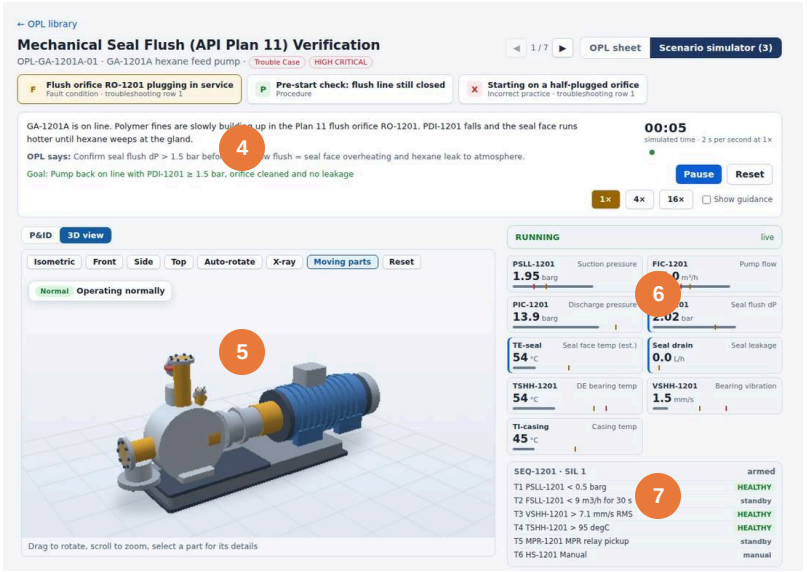
1,109 chunks | **447** typed facts | **624** relations | **564** linked records | **48** recorded conflicts

Source: Prototype Knowledge Studio, data quality page and knowledge graph, built from the case documents

A4 OPL library and scenario simulator



1. Standard OPL header with equipment, interlock, P&ID reference and classification.
2. Safety precautions, tools and detailed steps, each with an acceptance check.
3. Related scenarios to play, and links to the P&ID, interlock and datasheet.



4. The brief gives the symptom, the OPL rule and the goal.
5. A 3D machine reacts to the actions: it shakes, steams or stops.
6. Live readouts show alarm and trip marks for each instrument.
7. The interlock panel shows each trip as healthy, standby or tripped.

How the simulator works

Operator action
Valve move, start, reset or bypass

Process model
Pressures, flows, temperatures, levels

Set point reached?

Yes

Alarm or trip acts
A bypass is logged and nothing protects

Score
Goal held for a set time

56 OPL documents

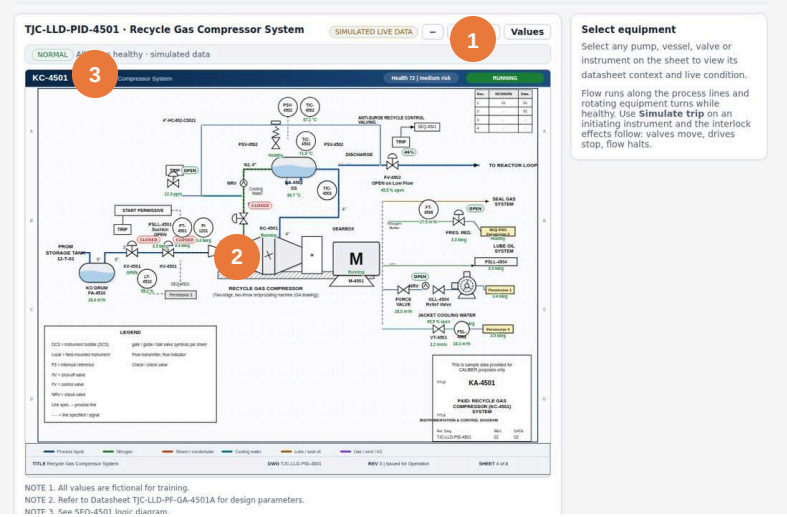
280 procedure steps

168 troubleshooting cases

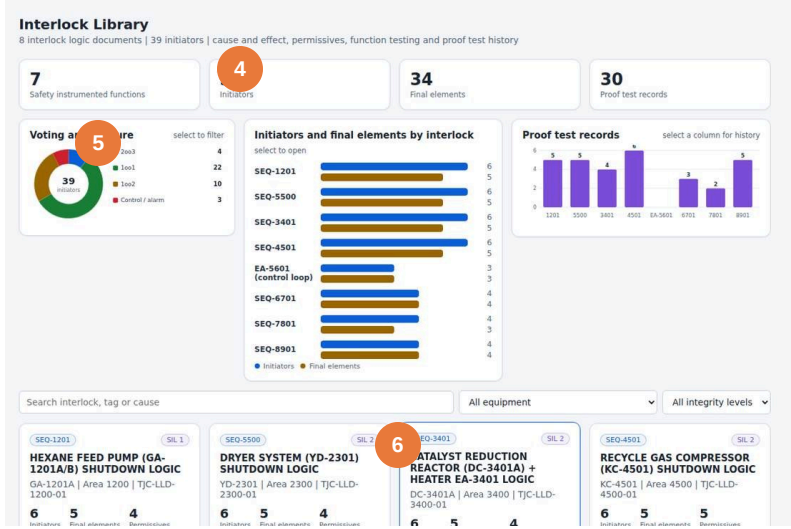
Score = 100, minus 20 per unsafe action, minus 12 per trip (up to 3), plus 2 per OPL step applied (up to 10). A failed run is capped at 35.

Source: Prototype OPL library and simulator on the 56 OPL documents of the case data. An OPL is a one point lesson

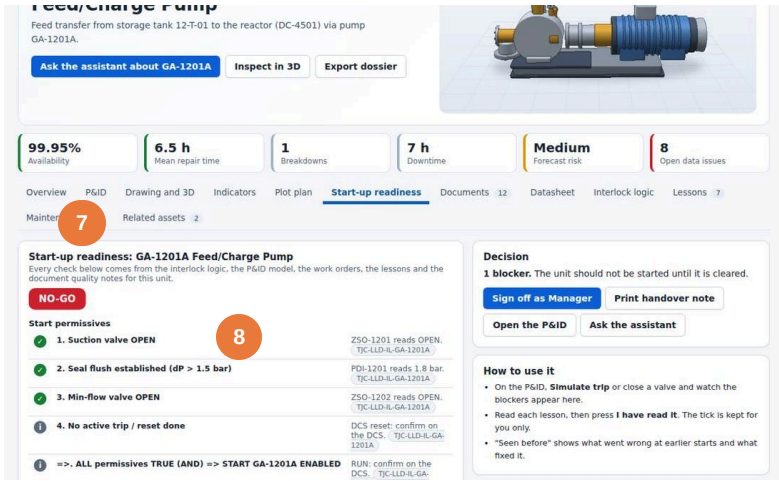
A5 Live P&ID, interlock library and start-up readiness



1. Each sheet shows simulated live data, with a Values toggle.
2. Valves open and close. Closing one stops the flow downstream and trips dependent permissives.
3. Simulate trip on an initiator and watch the interlock effects follow.



4. Eight interlock documents: 39 initiators, 34 final elements and 7 safety functions.
5. Voting architecture and proof test records at a glance.
6. Each interlock card links to its P&ID and equipment page.



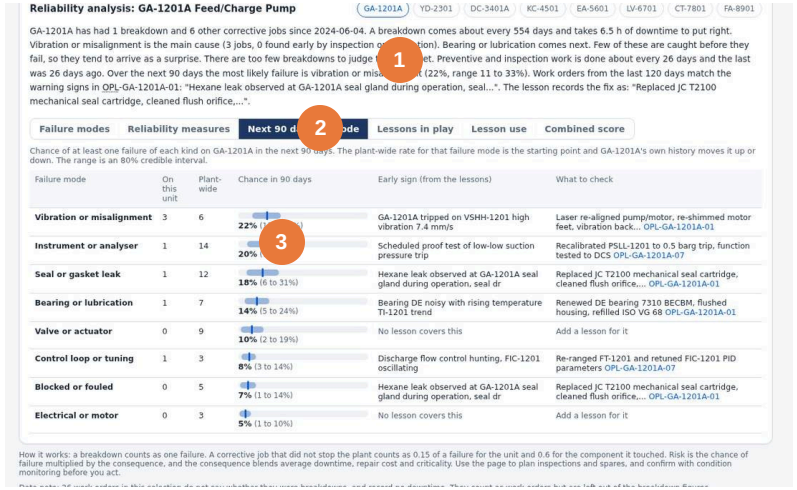
7. One decision per unit: GO, GO with checks or NO-GO.
8. Start permissives are read from the interlock logic and the live conditions.

8 P&IDs | **232** tagged items | **20** operable valves | **39** initiators | **37** permissives

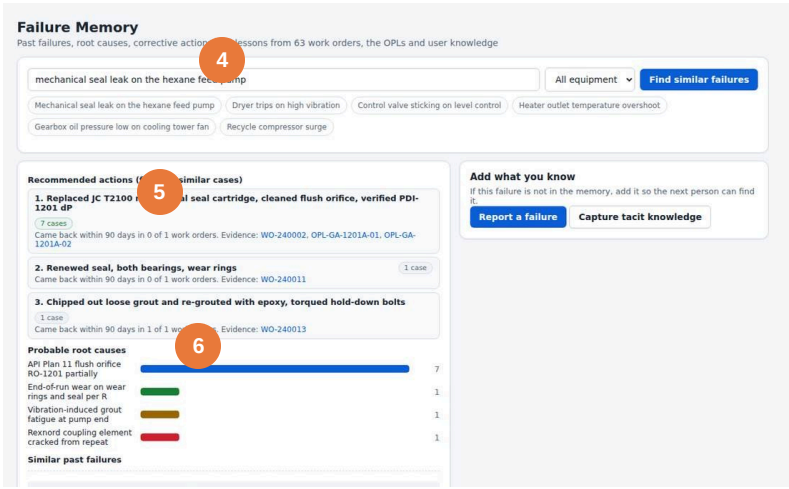
The same interlock data feeds the assistant, the OPL simulator, the scenario player and the readiness check, so the screens cannot disagree.

Source: Prototype P&ID viewer, interlock library and readiness page. Counts are read from the application

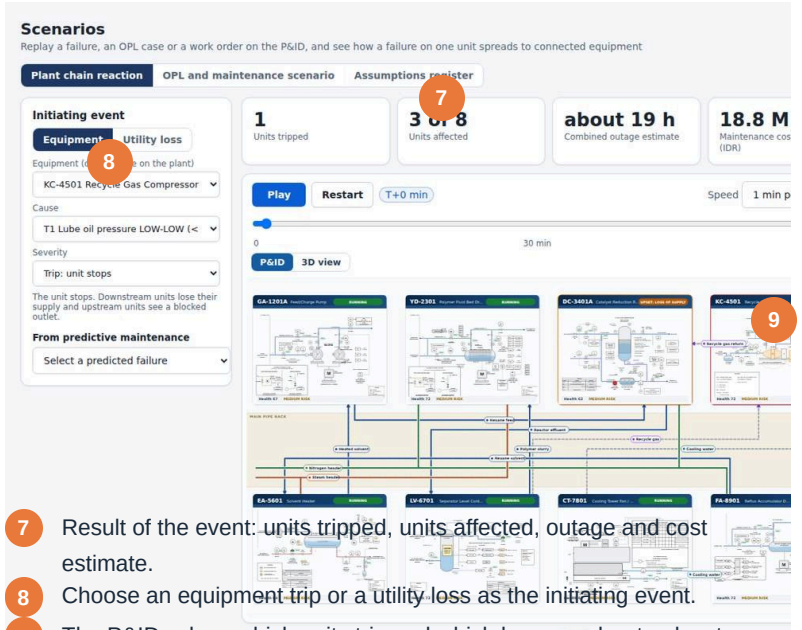
A6 Predictive maintenance, failure memory and chain reaction



1. A written summary of one unit: breakdowns, repair time, main cause and the next likely failure.
2. Tabs: failure modes, reliability measures, next 90 days, lessons and combined score.
3. Chance of failure in 90 days by mode, with an 80% range and the lesson that applies.



4. Type a symptom and optionally filter by equipment.
5. Recommended actions come from the most similar past cases, with how often each recurred.
6. Probable root causes are counted across the matches.



7. Result of the event: units tripped, units affected, outage and cost estimate.
8. Choose an equipment trip or a utility loss as the initiating event.
9. The P&IDs show which units trip and which lose supply, step by step.

Predictive page

Shows health, risk matrix, timeline and an intervention plan for 8 units and 42 components.

Failure memory

Matches a symptom to work orders, OPL cases and approved lessons, then ranks the actions.

Chain reaction

Replays how a trip or a utility loss spreads across the connected P&IDs.

211 work orders analysed

10 failure modes

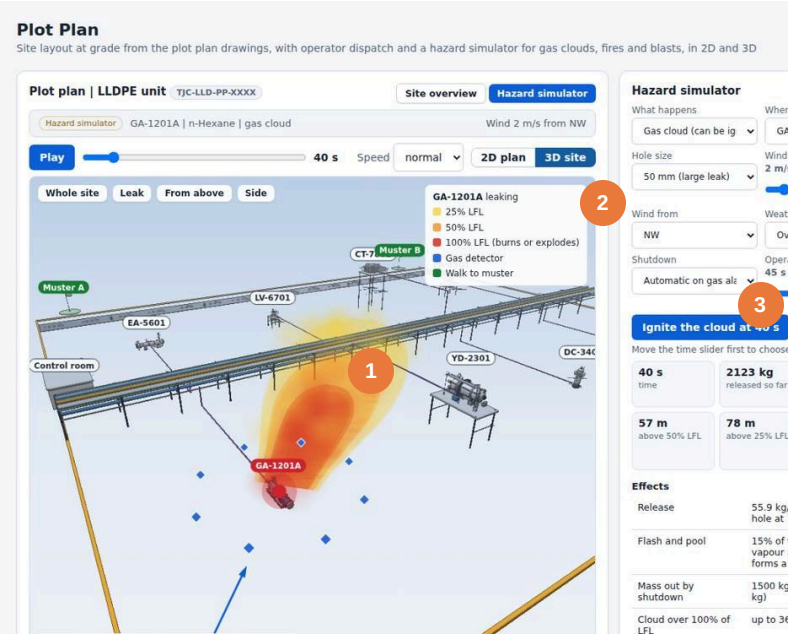
42 components

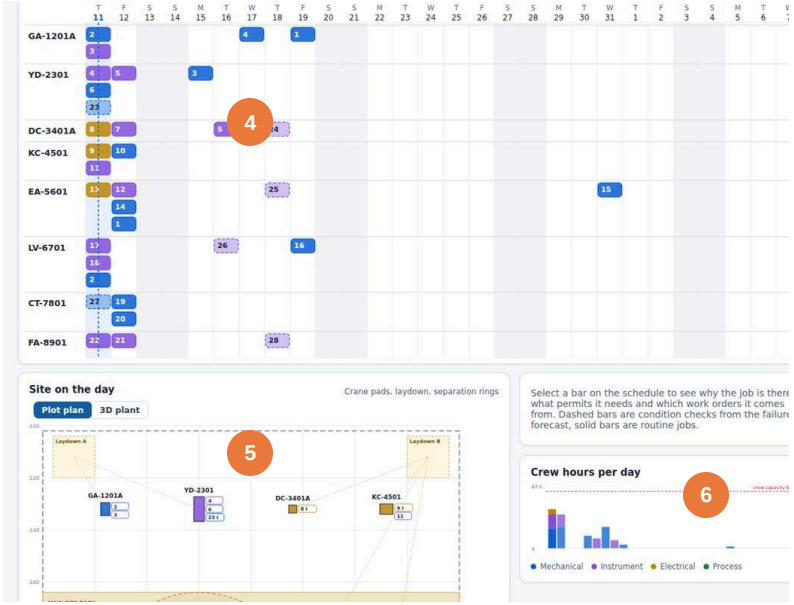
64 backtest forecasts

Limit: on this data a unit's own history does not predict which unit fails next (backtest AUC 0.47), so the combined score is advisory and the page ranks units by consequence.

Source: Prototype reliability pages computed in the browser from the 211 work orders and the OPL lessons

A7 Hazard simulator, maintenance site plan and plant replay







1. A gas cloud drifts with the wind over the true-scale plot plan, in 2D and 3D.
2. Zones show 25%, 50% and 100% of the lower flammable limit, with detectors and muster points.
3. Inputs are hole size, wind, weather, shutdown mode and operator reaction time.
4. A four-week schedule for each unit, built from routine intervals and the failure forecast.
5. The site on the day: crane pads, laydown areas and separation rings.
6. Crew hours per day against capacity.
7. The date line shows breakdowns, units in maintenance and the work order in progress.
8. Labels name the work order at each unit, and units in breakdown turn red.
9. Counters: completed work orders, breakdowns, downtime and availability.

Hazard models

Orifice release, flash and pool, Gaussian plume with transport delay, jet and pool fire, blast and muster walk.

Planner checks

Permits, hot work near open lines, one crane, crew capacity and confined space entry.

Replay

Steps through the history and stops inside each breakdown, so the right unit is shown.

Source: Prototype plot plan, hazard simulator, site planner and replay. Hazard results are screening estimates, not a consequence study

A8 Data model and storage

Three storage tiers

Browser

IndexedDB akp_kb

Chunk list and term index rebuilt on change. Works offline.

Shared store

Netlify Blobs through /api/kb

Collections uk, wo, fb, audit, qa. Last write wins; a base version in the request turns a newer copy into a conflict.

Production target

PostgreSQL 15 with pgvector

25 tables, generated tsvector, GIN indexes, HNSW cosine index on vector(1536) and row-level role filter.

Tables by group

Assets and documents

asset, component, document, document_version, chunk, fact, relation

Operations

interlock, interlock_trigger, instrument, opl, opl_case, work_order, failure_event

Knowledge and governance

knowledge_item, knowledge_revision, knowledge_review, qa_log, qa_citation, feedback, eval_case

Other

eval_run, audit_log, app_user, status_vocab (11 trust codes)

Anatomy of one chunk

chunk_id	il:TJC-LLD-IL-KC-4501#03
kind, record	il, interlock sheet
title, section	Interlock | KC-4501 | T3
text	VSHH-4505 80 um, 2oo3 ...
tags	{KC-4501, VSHH-4505}
status, trust	approved, 1.00
revision, date	Rev 3, 2025-02-11
acl roles	all roles
hash, tokens	cyrb53, 112
embedding	vector(1536), when connected

Indexes: GIN on the tsvector and on tags, B-tree on asset and status, HNSW (cosine) on the embedding. Embeddings are recomputed only when the content hash changes.

1,109 chunks | **447** typed facts | **624** relations | **564** linked records | **98** documents | **211** work orders

Source: Prototype data layer, Netlify Blobs function /api/kb and docs/DATA_MODEL.sql (25 tables)

A9 Retrieval, ranking and confidence algorithms

1 Question
Tags found by pattern; equipment inferred from names and aliases

2 Tokenise
Stemming, synonyms and plant stop words; tags kept whole

3 Score
BM25 over weighted fields; k1 1.4, b 0.75

4 Re-weight
Status trust, source kind, equipment scope and role fit

5 Fuse
With embeddings: 0.65 x BM25F + 0.35 x cosine x 40

6 Select
Role filter, at most 2 passages per document

Field weights in BM25F

Equipment tag	x 2.5
Title	x 2.0
Section heading	x 1.5
Body text	x 1.0

Why BM25F first: plant text is short and full of identifiers that exact matching handles well. The weights stop a long body from outranking a heading that names the tag.

Multipliers on the BM25F score

Status trust	0.35 to 1.00
Source-kind prior	0.55 to 1.05
Equipment in scope	x 1.45
Other equipment	x 0.70
Role fit	x 1.10 to 1.15

Kind priors range from 1.05 for interlock and OPL records down to 0.55 for data-quality notes.

Confidence score, 0 to 100

Question words found	45
Best match strength	22
Approval status	15
Source agreement	10
Number of sources	8

Caps apply after weighting: 60 for a broad question, 45 if under half the question words are found, 40 for different equipment, 50 if no single passage covers the question.

High: 70 and above | Medium: 45 to 69 | Low: below 45

Number and quote check. Each figure with a unit (barg, degC, mm/s, m3/h, kW, IDR or %) is looked up in the cited text. A figure that cannot be found is listed under the answer and the confidence drops. Calculated tools report their own confidence.

Source: Prototype retrieval index and assistant. Parameters are the values used in the running application

A10 Reliability algorithms and their validation

1 Classify
10 failure modes read from the problem, cause and action text

2 Risk priority
RPN = Severity x Occurrence x Detection

3 Forecast
Gamma-Poisson chance of failure, with an 80% range

4 Combine
Ridge logistic regression on 8 features

5 Validate
Rolling-origin backtest of 64 forecasts

Risk priority number

- Severity: percentile of downtime (55%) and cost (45%), scaled 1 to 10.
- Occurrence: 1 + 2.4 log2(1 + 3 x rate per year), at most 10.
- Detection: 1 + 9 x (1 - share found early), minus 1.2 if an OPL covers the mode.

Gamma-Poisson forecast

- Prior from the fleet rate: shape 1.5, rate 1.5 divided by the fleet rate.
- After n own events in T days: shape 1.5 + n, rate prior + T.
- Chance in h days = 1 - exp(-mean rate x h). Few events shrink toward the fleet rate.

Combined score

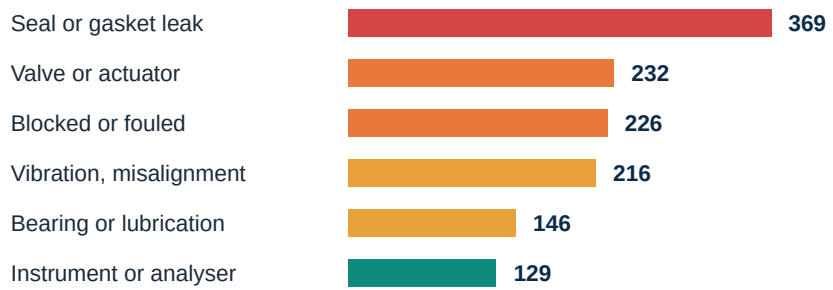
- Eight standardised features, such as recent breakdowns, trend, repeats and overdue preventive work.
- 300 gradient-descent iterations with an L2 penalty.
- The backtest trains on the past and forecasts the next 90 days, then advances 20 days.

64 forecasts tested

0.26 Brier score

0.47 AUC; 0.50 is chance

Failure modes by risk priority number



Limit stated in the product. On this data a unit's own history does not predict which unit fails next, so the page ranks units by consequence and shows the combined score as advisory. More history or live condition data would improve it, and the pilot is the place to collect them.

Source: Prototype reliability analytics on the 211 work orders; 63 breakdown and corrective jobs behind the failure-mode view